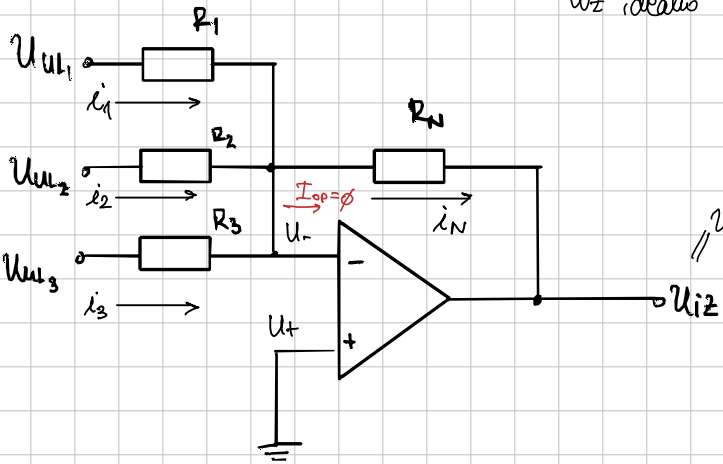
Sklop za zbrajanje



Uz idealno OP $\rightarrow$ $u_- = u_+ = 0$

$$i_1 + i_2 + i_3 = i_N$$

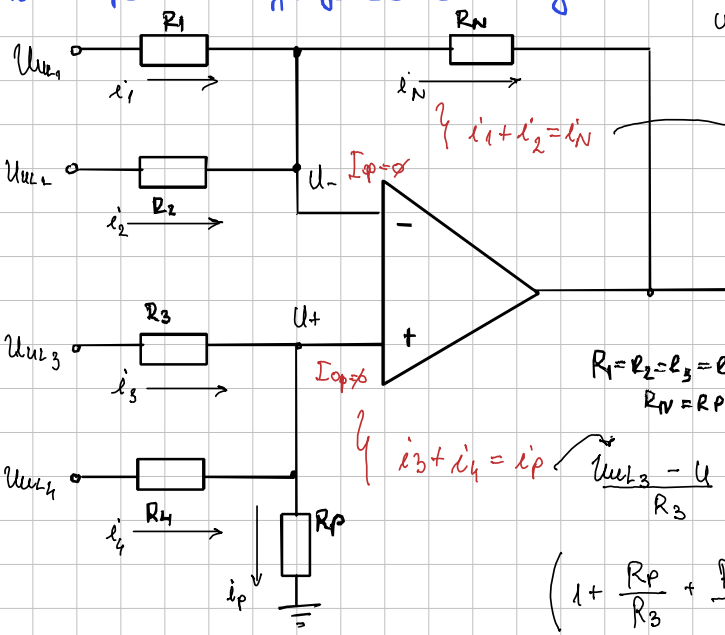
$$u_{iz} = -i_N \cdot R_N$$

$$\frac{u_{ul_1}}{R_1} + \frac{u_{ul_2}}{R_2} + \frac{u_{ul_3}}{R_3} = \frac{-u_{iz}}{R_N}$$

ako je $R_1 = R_2 = R_3$: $\rightarrow$ $u_{iz} = \frac{-R_N}{R_1}\left(u_{ul_1} + u_{ul_2} + u_{ul_3}\right)$

a ako je $R_1 = R_2 = R_3 = R_N$: $u_{iz} = -\left(u_{ul_1} + u_{ul_2} + u_{ul_3}\right)$

Sklop za zbrajanje i oduzimanje



Uz idealno OP: $u_+ = u_- = u$

$i_1 + i_2 = i_N$

$$\frac{u_{ul_1} - u}{R_1} + \frac{u_{ul_2} - u}{R_2} = \frac{u - u_{iz}}{R_N}$$

$$\hookrightarrow u_{iz} = \left(1 + \frac{R_N}{R_1} + \frac{R_N}{R_2}\right)u - \frac{R_2}{R_1}u_{ul_1} - \frac{R_N}{R_2}u_{ul_2}$$

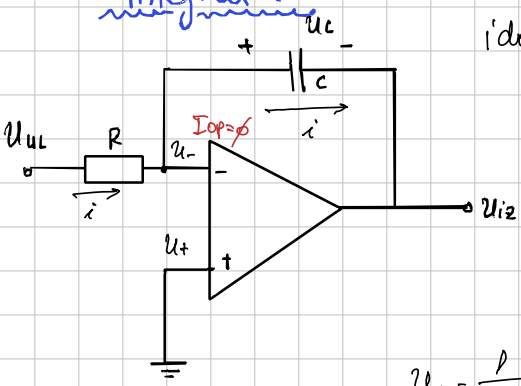
$R_1 = R_2 = R_3 = R_4$, $R_N = R_P$ $\Rightarrow$ $u_{iz} = \frac{R_N}{R_1}\left(-u_{ul_1} - u_{ul_2} + u_{ul_3} + u_{ul_4}\right)$

$i_3 + i_4 = i_P$

$$\frac{u_{ul_3} - u}{R_3} + \frac{u_{ul_4} - u}{R_4} = \frac{u}{R_P}$$

$$\left(1 + \frac{R_P}{R_3} + \frac{R_P}{R_4}\right)u = \frac{R_P}{R_3}u_{ul_3} + \frac{R_P}{R_4} \cdot u_{ul_4}$$

Integrator



idealno OP: $u_- = u_+ = 0$

$$i(t) = \frac{u_{ul}(t)}{R}$$

$$u_C(t) = \frac{1}{C}\int_0^t i(t)\,dt = \frac{1}{C}\int_0^t \frac{u_{ul}(t)}{R}\,dt + u_{C0}$$

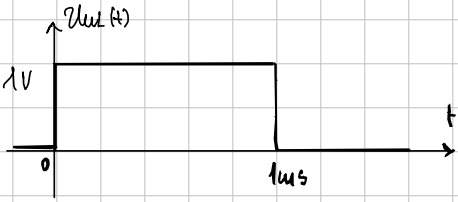
$$u_C = \frac{1}{RC}\int_0^t u_{ul}(t)\,dt + u_{C0}$$

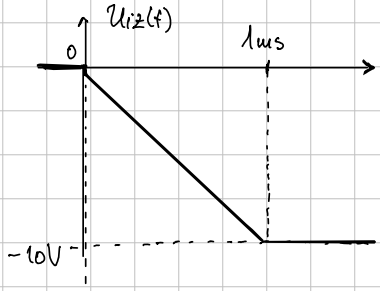
ako je $u_{C0} = u_C(t=0) \rightarrow u_{iz}(t) = -u_C(t) = -\left(\frac{1}{RC}\int_0^t u_{ul}(t)\,dt + u_{C0}\right)$

Primjer 10.5.) $u_{iz}(t)$ (Integrator ↑) ?

$u_{ul}(t)$ je impuls ; $U_{ul_m} = 1V$, $T_p = 1ms$ $\qquad U_{C0} = 0V$

$R = 10k\Omega$, $C = 10nF$



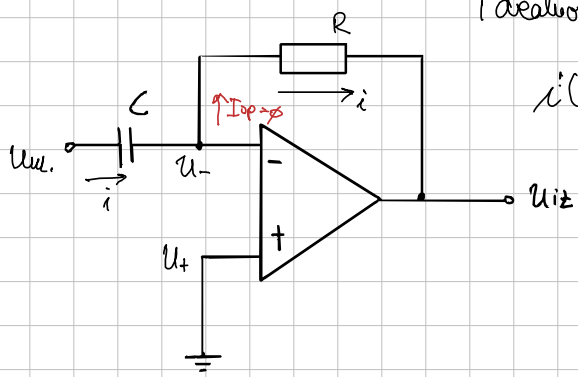


rješenje

$$u_{iz} = -u_C(t) = \frac{-1}{RC}\int_0^{T_p} u_{ul}(t)\cdot dt - \cancel{u_{C0}}^{\,0} = \frac{-1}{10^4\cdot 10^{-8}}\int_0^{1ms} \underbrace{u_{ul}(t)}_{U_{ul_1}=1}\,dt$$

$$u_{iz} = -\frac{t}{0.1m} = -\frac{1m}{0.1m} = \underline{\underline{-10V}}$$

Derivator:



idealno OP: $u_- = u_+ = 0$

$$i(t) = C\cdot\frac{d\,u_{ul}(t)}{dt}$$

$$u_{iz} = -R\cdot i = -R\cdot C\cdot\frac{d\,u_{ul}(t)}{dt}$$

Multivibratori

→ impulsni skokovi

- stanje niske ili stanje visoke razine
- Stabilno = stanje u kojem ostaje trajno
- kvazistabilno = ostaje ograničeno vrijeme

⟹ 3 vrste:

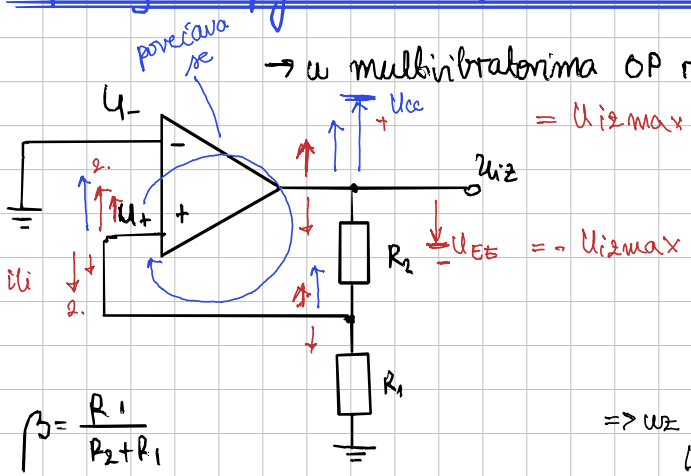
▷ bistabilni = bistabil → oba stabilna

▷ monostabilni = monostabil → jedno stabilno, drugo kvazistabilno

▷ astabilni = astabil → oba kvazistabilna

Operacijsko pojačalo s pozitivnom povratnom vezom

→ u multivibratorima OP rade s poz. pov. vezom



$$\beta = \frac{R_1}{R_2 + R_1}$$

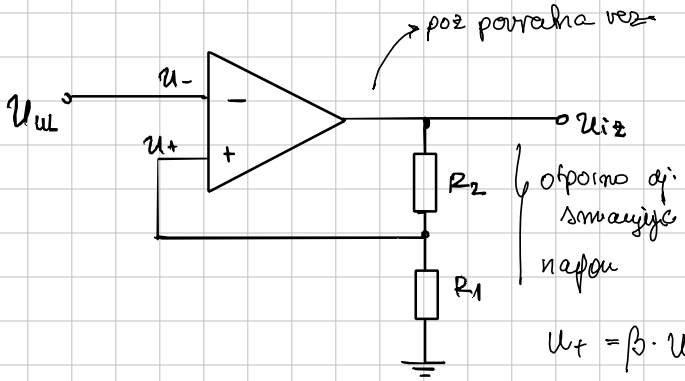
⟹ uz $\beta A_{vop} > 1$ izlazni napon je sve veći i na kraju postaje napon zasićenja U_{izmax}

⟹ uz neg smetnju na "+" ulazu
↳ izlazni napon zasićenja $U_{izmin} = -U_{izmax}$

⟹ Komparatori

Komparator napona

usporeduje dva ulazna napona

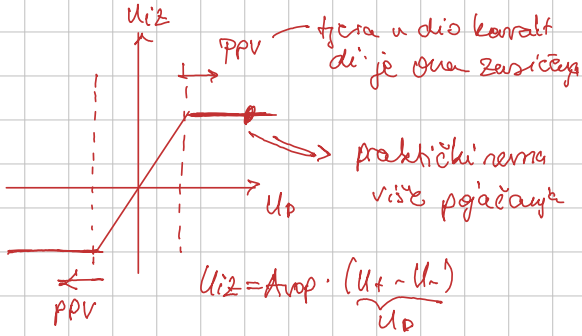


$$U_+ = \frac{R_1}{R_1+R_2} \cdot U_{iz} = \beta U_{iz}$$

$$U_+ = \beta \cdot U_{iz} > -U_{izmax} = U_-$$

$$\Rightarrow U_+ > U_- \Rightarrow U_{iz} = A_{vop} \cdot (U_+ - U_-) = U_{izmax}$$

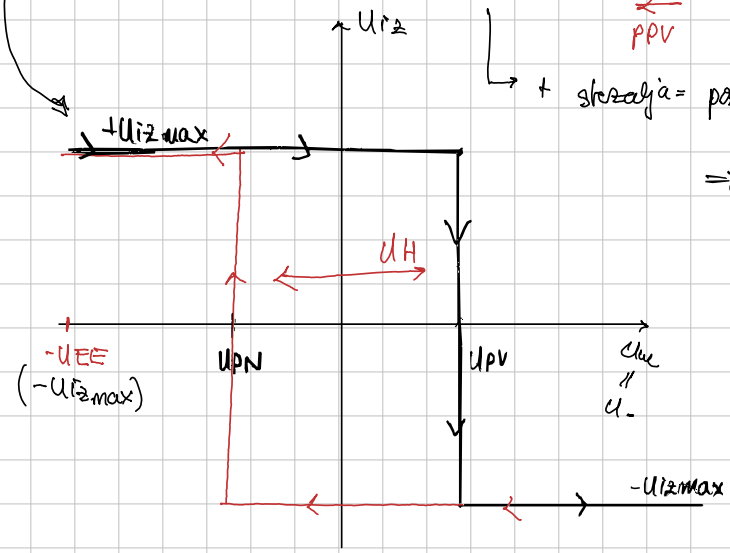
Formula vrijedi samo u linearnom podnožju, tj dijelu karakt gdje izlazni napon ovisi linearno o ulaznom. diferenc. napon



$\boxed{U_{PV} \rightarrow prag}$ (napon praga visoko)

$$U_+ = \beta \cdot U_{izmax} = \frac{R_1}{R_1+R_2}(+U_{izmax})$$

+ stezaljka = poz napon



$\Rightarrow U_{PV} \Rightarrow U_- > U_+$

$\Rightarrow U_{iz} = -U_{izmax}$

Kada minus postavimo na visoku razinu, napon koji vraćamo na + stezaljku neki negativni napon

$$U_+ = \beta \cdot (-U_{izmax}) = U_{PN}$$

↳ zbog razmaka sklop je puno manje osjetljiv na smetnje

- U_{PV} = prag okidanja za visoku razinu
- U_{PN} = —||— nisku razinu

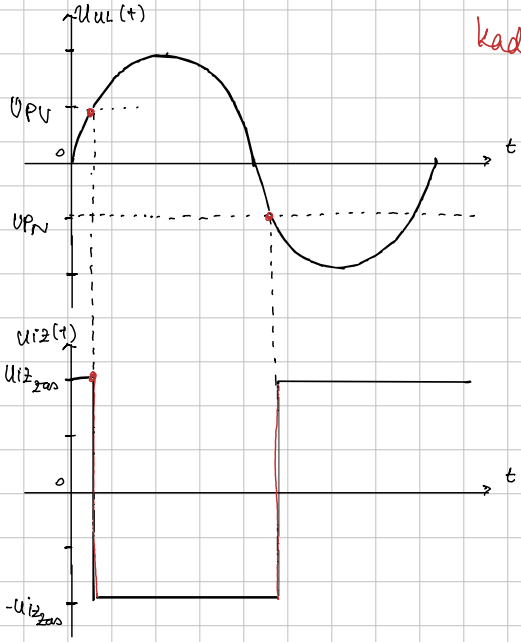
$$U_{PV} - U_{PN} = U_H$$

napon histereze

$$+\beta \cdot U_{izmax} - (-\beta U_{izmax}) = U_H$$

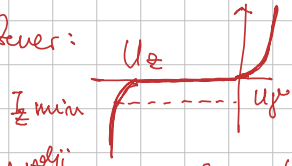
$$2\beta \cdot U_{izmax} = U_H$$

Odziv komparatora na sinusnu pobudu

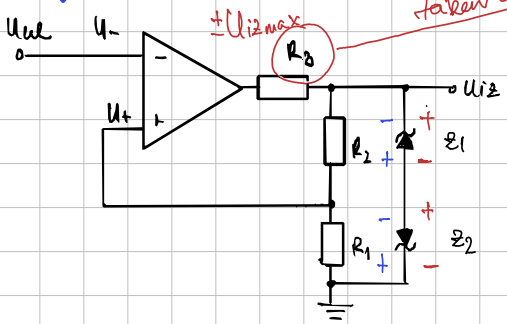


kada U_{ul} poraste za iznos U_{PV} dolazi do okidanja (promjena stanja na izlazu)

Zener:



Ograničavanje izlaznog napona



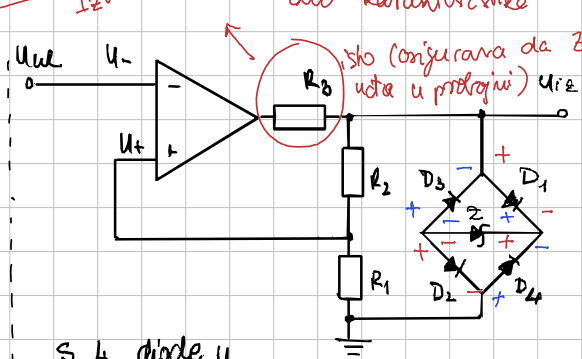
takav da se zadovolji $I_{Z\,min}$ i omogući prolazak u probojni dio karakteristike

s dvije Zenerove diode

$$U_{iz} = \pm (U_Z + U_D)$$

$+U_{izmax} \rightarrow$ Z_1 proboj (U_Z)
Z_2 propusno (U_γ ili U_D)

$-U_{izmax} \rightarrow$ Z_1 propusno (U_γ)
Z_2 u proboju (U_Z)



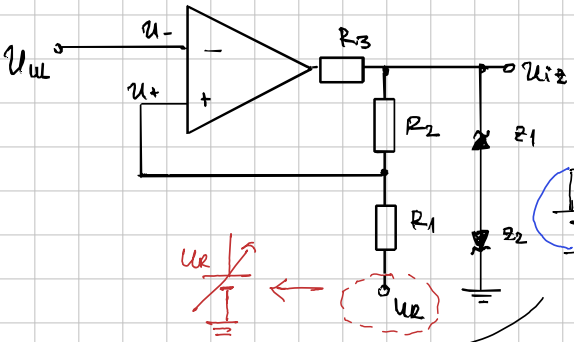
s 4 diode u mostnom spoju i s jednom Zenerovom diodom

$$U_{iz} = \pm (U_Z + 2U_D)$$

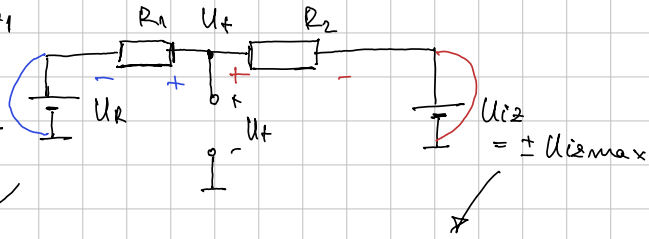
$+U_{izmax} \rightarrow$ D_1, D_2 propusno
Z u proboju

$-U_{izmax} \rightarrow$ D_3, D_4 propusno
Z u proboju

Komparator s pomaknutom prijenosnom karakt.



U_+ određuje prag okidanja



u_{iz} se mijenja ovisno o prijelazima okidanja komp.

SUPERPOZICIJA

$$U_+ = \underbrace{U_R \cdot \frac{R_2}{R_1+R_2}} + \underbrace{U_{iz} \cdot \frac{R_1}{R_1+R_2}}$$

$$U_{PV} = U_R \cdot \frac{R_2}{R_1+R_2} + U_{izmax} \frac{R_1}{R_1+R_2}$$

$$U_{PN} = U_R \cdot \frac{R_2}{R_1+R_2} - U_{izmax} \cdot \frac{R_1}{R_1+R_2}$$

$$U_H = U_{PV} - U_{PN} = 2U_{izmax} \cdot \frac{R_1}{R_1+R_2}$$ Širina histereze

$$U_{CH} = \frac{U_{PV}+U_{PN}}{2} = U_R \frac{R_2}{R_1+R_2}$$

Primjer 10.6: $R_1 = 1.5\,k\Omega$ $\underbrace{U_Z = 4.3V \quad U_D = 0.7V}_{U_{izmax} = \pm 5V = \pm(U_Z+U_D)}$

$U_H = 0.1V$

$U_{CH} = 1V$

$$U_H = 2\,U_{izmax} \cdot \frac{R_1}{R_1+R_2} \overset{?}{=}$$

$$U_{CH} = U_R \cdot \frac{R_2}{R_1+R_2} \overset{?}{=}$$

$$2U_{izmax} = \left(1+\frac{R_2}{R_1}\right)U_H$$

$$R_2 = \left[\frac{2U_{izmax}}{U_H} - 1\right] \cdot R_1 = 14.9\,k\Omega$$

$$U_R = U_{CH}\left(1+\frac{R_1}{R_2}\right) = 1.01V$$